① 4x1024 mem module $= 2^{12}$

20 bit address port

8 bit data port

$A_{11} - A_0 \rightarrow$ memory module (offset)

$A_{19} - A_{12} \rightarrow$ chip select logic (decoding)

80000H + 1000 − 1 = 80FFFH → ending

$A_{19} = 1$ $A_{18} - A_{12} = 0 \Rightarrow$ logic gate that outputs active (low) when $A_{19}=1$ and $A_{18} - A_{12} = 0$

2X1024 memory module

4X1024 X 2 byte word

24 bit address port

16 bit data port

B00000 H

1011 0000 0000 0000 0000 0000

1011 0000 0000 1111 1111 1111

16 bit data bus but modules are 8-bit ⇒ we need to connect two modules in parallel ($D_0 - D_7$ & $D_8 - D_{15}$)

for 16 bit word ⟶ Bank of size 2K words

⇒ we need 2 banks (each bank → 2 chips)

$\frac{4096}{2}$ words ⇒ 11 address lines $A_1 \rightarrow A_{11}$

A_0 select the byte

A_{12} selects blw bank 1 or 2

$A_{13} - A_{23} \rightarrow$ chip select

$A_{23}, A_{21}, A_{20} = 1$

$A_{22}, A_{19} - A_{13} = 0$

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16 bit address port

8 bit data port

1024 byte mem module ⇒ $A_0 - A_9$ is used for offset

A_{15} AND A_{14}: module is selected when $A_{14}=1$ & $A_{15}=1$

$A_{13} - A_{10}$ → unused

Zeroth address ⇒ $A_9 - A_0 = 0000 \ldots 0$

$11 \underbrace{xx\ xx}_{2^4} 00\ 0000\ 0000$ → 16 addresses

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4 low order address lines → module select

6 high order " " → mem module

16 modules, T → access time

$T_{eff} = T/m \times S$ for simple interleaving

$T_{eff} = T/m \times gcd(m, S)$ for complex "

S → skip distance

a) $T/16 \times 2 = T/8$ b) $T/16 \times 4 = T/4$ c) $T/16 \times 5 = 5T/16$ d) $T/16 \times 1 = \frac{T}{16}$

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h+n+l address lines

2^n size mem modules in 2D array 2^h columns 2^l rows

low order l-lines are to generate 2^l row select lines

h lines → selects column → $0 - 2^h - 1$ l-lines → selects row → $0 - 2^l - 1$

$A = j \times 2^{n+l} + k \times 2^{l} + i \qquad k \in [0, 2^{n})$

Setup / notation (from the problem).

- There are $h + n + l$ address lines total. Label address bits so the least-significant bit is A_0.
- There are 2^h columns (indexed by $j = 0, \ldots, 2^h - 1$). The high h address lines select the column.
- There are 2^l rows (indexed by $i = 0, \ldots, 2^l - 1$). The low l address lines select the row.
- Each memory module stores 2^n locations: the middle n address lines go to the internal address inputs of every module to select one of those 2^n locations. Let $m \in \{0, \ldots, 2^n - 1\}$ be the module-internal address (the middle n bits).
- So an address has the bit layout (MSB → LSB):

$$[\text{column } (h \text{ bits})]\ [\text{module offset } (n \text{ bits})]\ [\text{row } (l \text{ bits})].$$

Addresses that map to module (i, j).
Fix a row i and column j. The module (i, j) is selected when the low l bits equal i and the high h bits equal j; the middle n bits m pick the location inside that module. Thus every physical address that maps into module (i, j) has the binary form

$$(j \,\|\, m \,\|\, i),$$

where "$\|$" denotes bit concatenation. j is h bits, m is n bits, and i is l bits.

In arithmetic form (with bit-shifts).

$$\boxed{\text{Address}(i, j, m) = j \cdot 2^{n+l} + m \cdot 2^{l} + i, \quad m = 0, 1, \ldots, 2^n - 1.}$$

Equivalently in bitwise notation:

$$\text{Address}(i, j, m) = (j \ll (n + l)) \mid (m \ll l) \mid i.$$

Address range for module (i, j).
As m runs 0 to $2^n - 1$, the set of addresses mapped to (i, j) is

$$\{ j2^{n+l} + i + k \cdot 2^{l} : k = 0, 1, \ldots, 2^n - 1 \}.$$

So they are 2^n addresses spaced by 2^l (i.e. differ in the middle n bits) and all share the same low l bits $= i$ and high h bits $= j$.